Matgeo-q.2.5.17

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Question

Find a unit vector perpendicular to both the vectors **a** and **b**, where

$$\mathbf{a} = \hat{i} - 7\hat{j} + 7\hat{k} \tag{1}$$

$$\mathbf{b} = 3\hat{i} - 2\hat{j} + 2\hat{k} \tag{2}$$

Solution

Step 1: Strategy

A vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ is given by their cross product, $\mathbf{c} = \mathbf{a} \times \mathbf{b}$. To find the unit vector, we will first compute the cross product and then normalize it by dividing by its magnitude.

$$\hat{n} = \frac{\mathbf{a} \times \mathbf{b}}{\|\mathbf{a} \times \mathbf{b}\|} \tag{3}$$

Step 2: Computing the Cross Product

The cross product can be calculated using the determinant of a matrix:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -7 & 7 \\ 3 & -2 & 2 \end{vmatrix} \tag{4}$$

Expanding the determinant from (4):

$$\begin{aligned}
\mathbf{a} \times \mathbf{b} &= \hat{i} \begin{vmatrix} -7 & 7 \\ -2 & 2 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 7 \\ 3 & 2 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & -7 \\ 3 & -2 \end{vmatrix} \\
&= \hat{i}\,((-7)(2) - (7)(-2)) - \hat{j}\,((1)(2) - (7)(3)) + \hat{k}\,((1)(-2) - (-7)(3)) \\
&= \hat{i}\,(-14 + 14) - \hat{j}\,(2 - 21) + \hat{k}\,(-2 + 21) \\
&= 0\hat{i} - (-19)\,\hat{j} + 19\hat{k} \\
&= 19\hat{j} + 19\hat{k} \qquad (5)
\end{aligned}$$

Let's call this resulting vector $\mathbf{c} = 19\hat{j} + 19\hat{k}$.

Step 3: Finding the Magnitude

Now, we compute the magnitude of the cross product vector **c**:

$$\begin{aligned} \|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{c}\| &= \sqrt{(0)^2 + (19)^2 + (19)^2} \\ &= \sqrt{2 \times 19^2} \\ &= 19\sqrt{2} \end{aligned} \tag{6}$$

Step 4: Normalizing to get the Unit Vector

Finally, we find the unit vector $\hat{n}$ by dividing the vector **c** by its magnitude:

$$\begin{aligned} \hat{n} = \frac{\mathbf{a} \times \mathbf{b}}{\|\mathbf{a} \times \mathbf{b}\|} &= \frac{19\hat{j} + 19\hat{k}}{19\sqrt{2}} \\ &= \frac{19(\hat{j} + \hat{k})}{19\sqrt{2}} \\ &= \frac{1}{\sqrt{2}}\left(\hat{j} + \hat{k}\right) \end{aligned} \tag{7}$$

Final Answer

The unit vector perpendicular to both **a** and **b** is:

$$\hat{n} = \frac{1}{\sqrt{2}} \left(\hat{j} + \hat{k} \right)$$

or in component form:

$$\hat{n} = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \tag{8}$$

Graphical Representation

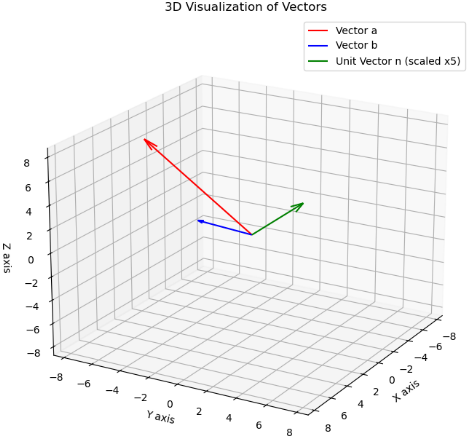

Figure: Vectors **a**, **b**, and the perpendicular unit vector $\hat{n}$.